

CARGOMULE · DESIGN DECISION RECORD

CargoMule TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	CGM-DDR-001	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/cargomule		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D10; items O1 and O2 remain proposed

■ Context

The TRL 2 review note (docs/REVIEW.md , session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish", and the design precis CGM-PRC-001 v0.2 listed further key design choices, each with options. On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided in favor of that recommendation. Items without a recommendation stay open. The same instruction approved three portfolio-wide decisions: SwapCell interface v0.3 adds a wake method for hosts without CAN, a charge-while-discharging mode and a latch vibration rating for vehicles; shared SwapCell packs are priced once and excluded from each dependent kit budget; and community designs pick co-design partners per area later.

■ Options considered

The options for each item are those listed in docs/REVIEW.md (session 2026-09-25, /populate) and in CGM-PRC-001 v0.2, Key design choices.

■ Decision

Table 1. Decided items.

#	Item	Decision
D1	Budget	Cut cost first (salvaged or budget 20 in idler wheel, budget disc calipers, a smaller and lighter enclosure), then raise <code>budget_usd</code> to \$1,000 because the cuts alone do not reach \$900. The redefined budget is carried into R12 as a \$1,000 target. Decided by Amish, 2026-09-25: go with recommendation.
D2	Battery	Option A: a 12S 36 V class LiFePO4 pack, 384 Wh, rather than a SwapCell pack. A SwapCell receiver stays a later fleet variant. Decided by Amish, 2026-09-25: go with recommendation.
D3	Frame material	Welded mild steel for the first prototype, with R8 relaxed from 35 kg to 40 kg. Decided by Amish, 2026-09-25: go with recommendation.
D4	Drive layout	Option A: one 250 W geared hub motor in the left wheel, with a plain idler wheel on the right. Decided by Amish, 2026-09-25: go with recommendation.
D5	Control law	Proportional assist with a rider-selectable fixed gain of 2 to 6, default $G = 4$, rather than a closed loop to a set tension. Decided by Amish, 2026-09-25: go with recommendation.
D6	Brakes	Option A: a mechanical overrun coupler on disc brakes on both wheels, plus an electronic assist cut, rather than an electrically actuated brake or a cable from the bike's lever. Decided by Amish, 2026-09-25: go with recommendation.
D7	Pitch wording	Soften "fits any bicycle" to "fits most bicycles" and "almost no added load" to "a fraction of the load" in <code>project.yaml</code> and <code>README.md</code> . The problem line is unchanged. Decided by Amish, 2026-09-25: go with recommendation.
D8	Force sensing	Sense pull force with a load cell at the drawbar (pitch-level choice kept), rather than a crank sensor on the bike or wheel-speed matching. Decided by Amish, 2026-09-25: go with recommendation.
D9	Wheels and hitch side	20 in (ETRTO 406) wheels and a hitch on the bicycle's left axle end, as proposed. Decided by Amish, 2026-09-25: go with recommendation.
D10	Handcart mode	Stays out of scope for now. Decided by Amish, 2026-09-25: go with recommendation.

Table 2. Items that remain open (no recommendation was made).

#	Item	Status
01	First users and region for co-design: market traders, a trades cooperative or a community food bank	Proposed, awaiting Amish (co-design partners to be picked per area later, as Amish directed for community designs)
02	Legal status of a motorized bicycle trailer on public roads and bike lanes in the first target country (EU pedal-assist exclusion, US federal and state rules)	Proposed, awaiting Amish; no recommendation was made and none is made here

■ Consequences

- `project.yaml`: `budget_usd` is \$1,000 and the pitch reads "fits most bicycles" and "a fraction of the load". `README.md` matches.
- CGM-PRB-001, CGM-PRC-001 and CGM-REQ-001 are revised to v0.3. R8 is relaxed to 40 kg and R12 is redefined as \$1,000 (the budget). No other requirement target changes. The key design choices in the precis are no longer "proposed".

- The BOM carries the D1 cost cuts: idler wheel \$30 (was \$55), disc calipers \$20 each (were \$30) and a 0.8 mm, 320 x 300 x 170 mm enclosure at \$25 (was \$30). The TRL 3 total is \$969 (CGM-CAL-001, I1).
- The SwapCell fleet variant under D2 is not designed at TRL 3. If it is taken up, it would build to SwapCell interface v0.3 (wake without CAN, charge while discharging, vehicle latch vibration rating), and the shared pack would be priced once in the SwapCell repo and left out of the CargoMule kit budget.
- TRL 3 calculations (CGM-CAL-001) show that R8 is still not met at 40 kg with the steel frame (about 43 kg). A response is proposed in `docs/REVIEW.md` and awaits Amish; this record does not decide it.
- TRL 4 is on hold by Amish's instruction. Nothing in this record authorizes building or testing.